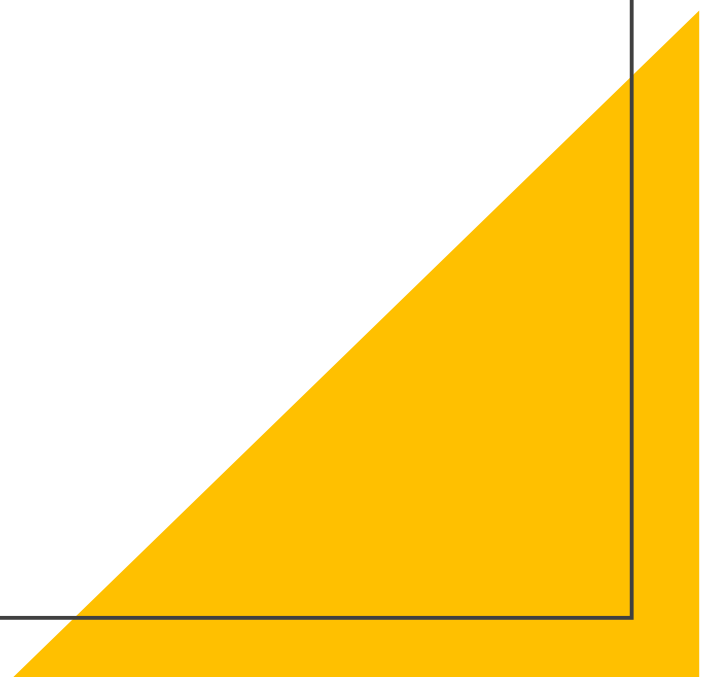


Getting started in Research and Analytics

Session 4: Survey Research Methods





Measurement and Assessment

In this session,
we'll cover scales
and factor
analysis.

We can answer the
question, "Is my
factor reliable and
valid?"



Why should we learn this?

To create and get the most out of our surveys.

To critically review existing surveys (e.g., HCAHPS).

A photograph of a wooden desk with a calculator, a pair of glasses, and a document. The document features a bar chart, a line graph, and two tables of data. A black pen is resting on the document. The calculator is a silver and black handheld device. The glasses are black-rimmed. The document is white with various colored charts and tables.

What is measurement and assessment?

- A different field within statistics, basis of surveys.
- We develop a measure and assess its accuracy in measuring the phenomena we are interested in (i.e., we assess our measure).
- **Reliability**: How reliable our measure is in producing consistent values of our factor (e.g., Cronbach's Alpha).
- **Validity**: How valid is the factor a measure of the phenomena we are interested in. In other words, is it measuring what it is supposed to measure (e.g., Confirmatory Factor Analysis). Content, construct, or criterion validity may be sufficient, consult clinical experts for content validity.

Types of measurement examples in surveys

- Behavior and beliefs (e.g., risky health behaviors, political beliefs).
- Cognitive aptitude and achievement (e.g., SATs).
- Assessment of mental illness (e.g., borderline personality disorder).
- FBI suspect profiling (e.g., psychometrics).
- Data reduction: Combining highly correlated variables into a single factor (e.g., combining heart rate, mean blood pressure, and age to represent a factor of “health status”).
- Use survey responses of multiple items to form single factors on beliefs of “Evidence Based Practices” in Nursing (e.g., 5-point Likert scale).
- We measure intangible beliefs and opinions. Unlike blood tests, we can't directly measure or test thoughts.

Analysis examples

- Cronbach's Alpha of a 5-point Likert scale ("Strongly Disagree" to "Strongly Agree") to assess reliability.
- Factor Analysis of multiple survey question to identify which variables are tapping into the same behavior or belief (i.e., Exploratory Factor Analysis).
- Confirmatory Factor Analysis to validate that specific variables tap into the same dimension.

Scales

Scales measure the level of behaviors or beliefs.



Scales are often used in survey research.

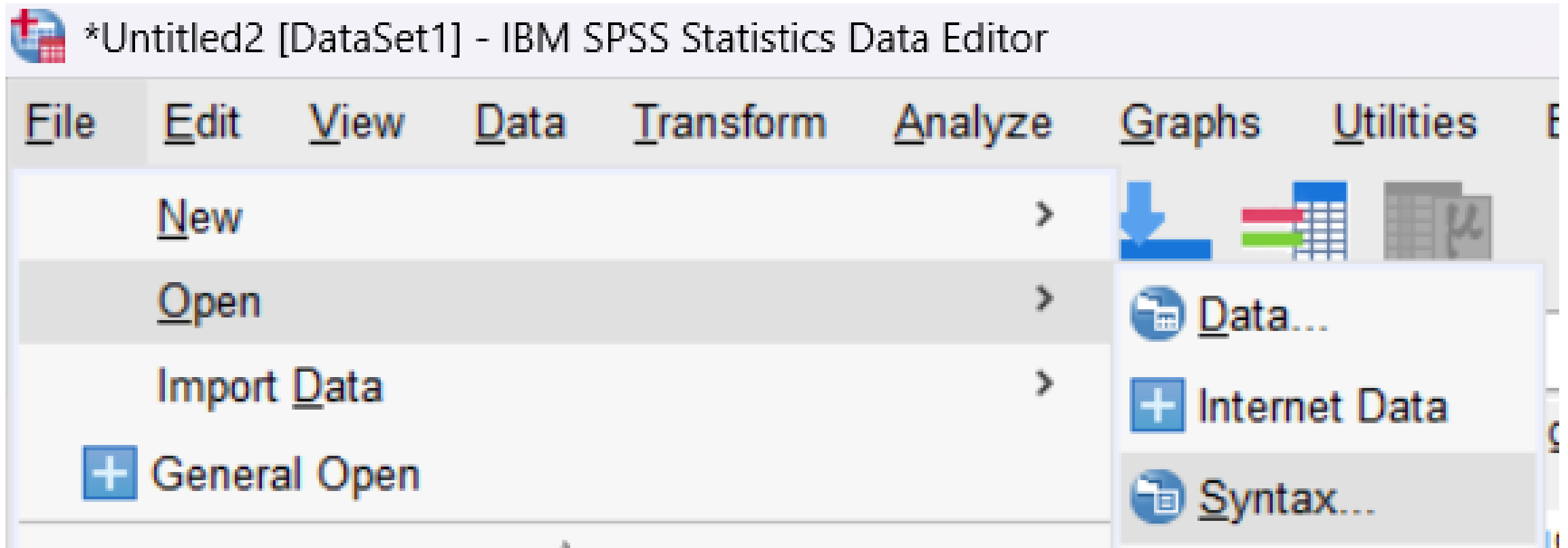


Common scales are 5-point Likert scales (e.g., “strongly agree”) but can be check lists like Guttman scales.



A common method for assessing Likert scales is the Cronbach’s Alpha which is an extension of correlations.

SPSS allows us to write time-saving code by opening a syntax file



Recoding variables in SPSS

Import the ebp.csv file, run code in a syntax file, and review the recoded data, see ebp.sps

* Recode new variables

```
RECODE q0002 q0006 q0008 q0013 q0015 q0026  
q0027 q0028
```

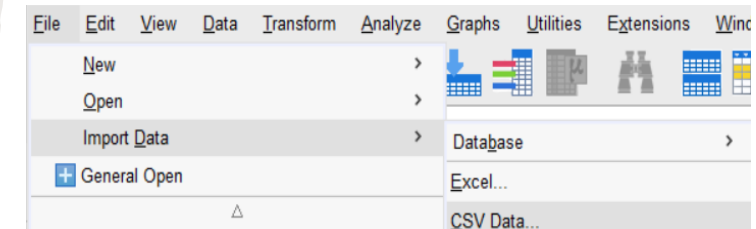
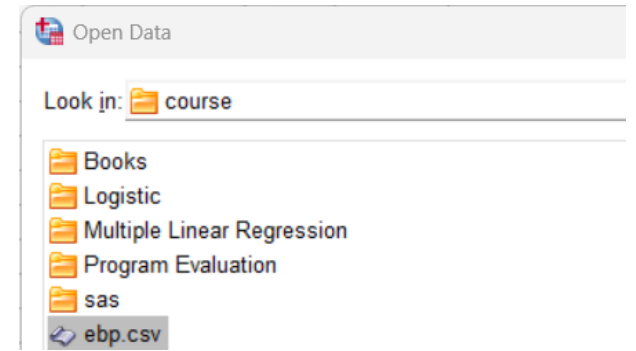
('Strongly Agree' = 5) ('Agree' = 4) ('Uncertain' = 3)
('Disagree' = 2)

('Strongly Disagree' = 1) ('NA' = SYSMIS)

```
INTO q0002r q0006r q0008r q0013r q0015r q0026r  
q0027r q0028r.
```

```
EXECUTE.
```

Scales need to be “written in the same direction” when we use the data (i.e., higher values represent more of the quality). But we sometimes write questions in reverse to make sure respondents are paying attention, why we need to recode variables.



	q0002	q0006	q0008
1	Agree	Agree	Agree
2	Strongly Agree	Strongly Agree	Strongly Agree
3	Strongly Agree	Agree	Agree

	q0002r	q0006r	q0008r
	4.00	4.00	4.00
	5.00	5.00	5.00
	5.00	4.00	4.00

Cronbach's alpha for 3 survey questions

Reliability Statistics

Cronbach's Alpha	N of Items
.896	3

Item Statistics

	Mean	Std. Deviation	N
q0026r	2.8554	1.19176	1141
q0027r	3.1946	1.20190	1141
q0028r	2.9921	1.15049	1141

Cronbach's Alpha assesses the reliability of multiple items in measuring the same dimension.

Alpha ranges from 0 to 1.

Common thresholds for high, moderate, and low reliability are 0.80, 0.60, and <0.60.

To the left, we see the Alpha level is quite high, 0.90. This indicates that the 3 measures reliably measure "Data Collection".

Optional interpretations in R's ham package

```
Factor2 <- alpha(items=c("q0026r", "q0027r",  
"q0028r"), data=ebp)  
interpret(Factor2)
```

Interpretations: Alpha

Your 3 item scale has a Cronbach's alpha of 0.896. This is generally considered as being in the 'very good' range.

The scale mean is 3.01 and has a standard deviation of 1.08.

Removing one of these item(s): , can improve the Cronbach's alpha in a new scale to a higher level than the current alpha based on all items.

40 row(s) of data excluded from the analysis because of missing data.

Example:
HCAHPS

The Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS)

```
graph TD; A[The Hospital Consumer Assessment of Healthcare Providers and Systems (HCAHPS)] --> B[The HCAHPS measures patient perspectives of their hospital care experience.]; B --> C[Because one question isn't enough, a factor analysis will help us know how well our questions work together.];
```

The HCAHPS measures patient perspectives of their hospital care experience.

Because one question isn't enough, a factor analysis will help us know how well our questions work together.

2015 HCAHPS Codebook

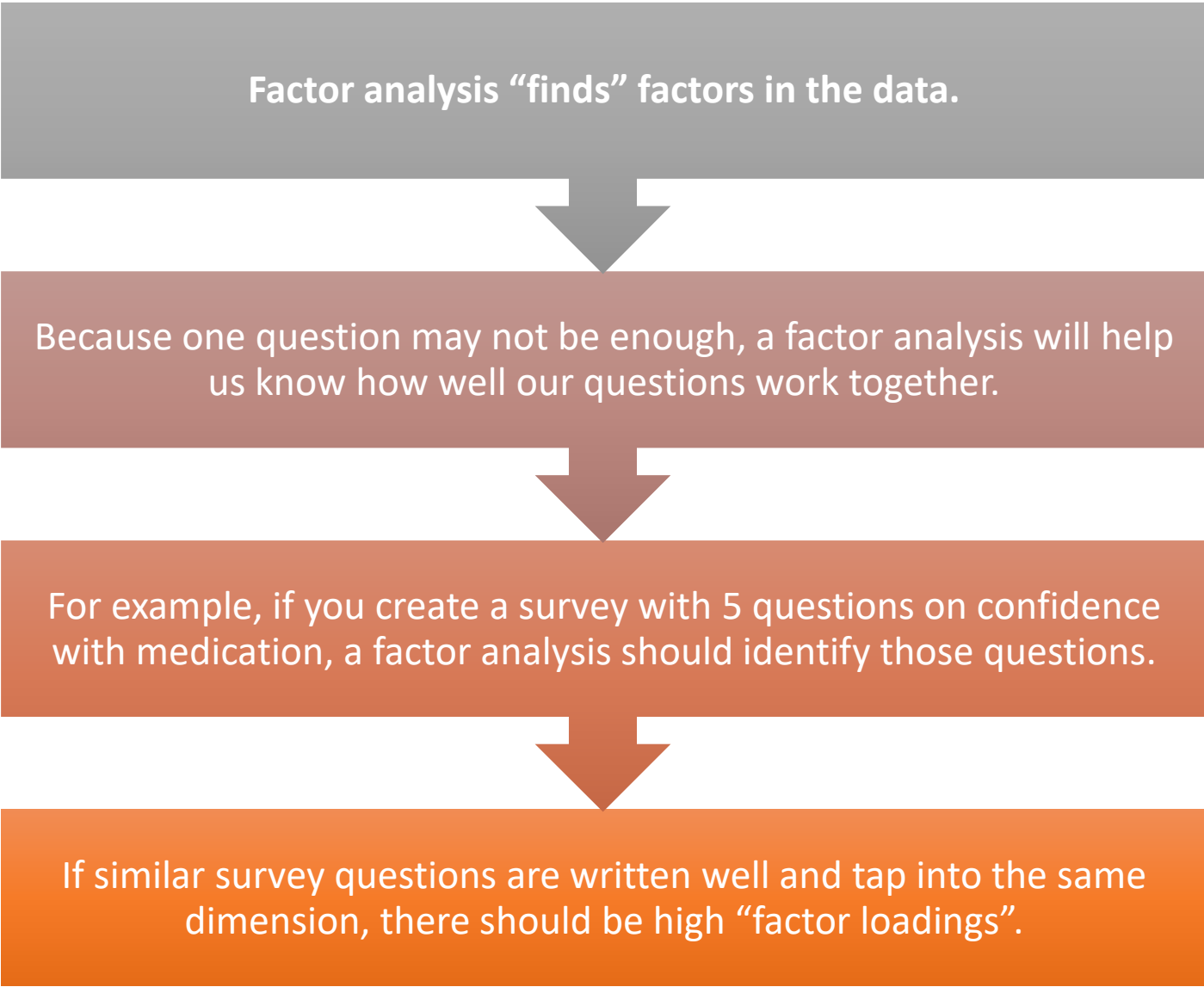
Alpha	Variable name	Category	Definition
N/A	i1	Global	H01.0 Rate hospital (9-10) 157019
	i2	Global	H02.0 Would recommend hospital 156654
0.83	i3	Nurse communication	H03.1 Treated w/courtesy/respect by Nurses (Always) 159708
	i4	Nurse communication	H03.2 Nurses listened carefully to you (Always) 159241
	i5	Nurse communication	H03.3 Nurses explained things understandably (Always) 158893
0.86	i6	Doctor communication	H04.1 Treated w/courtesy/respect by Drs (Always) 159590
	i7	Doctor communication	H04.2 Drs listened carefully to you (Always) 158478
	i8	Doctor communication	H04.3 Drs explained things understandably (Always) 156802
0.43	i9	Cleanliness and quiet	H05.1 Room and bathroom kept clean during stay (Always) 158392
	i10	Cleanliness and quiet	H05.2 Quiet around room at night (Always) 159053
0.65	i11	Responsiveness of hospital staff	H06.1 Got help as soon as wanted after you pressed the call button (Always) 138283
	i12	Responsiveness of hospital staff	H06.2 Help going to bathroom as soon as wanted (Always) 91498
0.77	i13	Pain Management	H07.1 Pain well controlled during stay (Always) 111066
	i14	Pain Management	H07.2 Did everything to help your pain (Always) 110953
0.63	i15	Communication about medicines	H08.1 Told what medicine was for (Always) 92177
	i16	Communication about medicines	H08.2 Staff described med side effects (Always) 91409
N/A	i17y	Discharge information	H09.1 Talked about help you would need : Yes 146412
N/A	i18y	Discharge information	H09.2 Received info re: symptoms to look for : Yes 145358
0.78	i19	Care Transition	H10.1 Preferences taken into account in deciding health care needs when I left (Strongly Agree) 153442
	i20	Care Transition	H10.2 Good understanding of things responsible for in managing my health (Strongly Agree) 155574
	i21	Care Transition	H10.3 Clearly understood purpose for taking each of my medications (Strongly Agree) 141772
0.91	Overall		
N/A	Age, Length of stay, Gender, Medical Center, Service line		

Removing i21 from “Care Transition”, can improve Cronbach’s Alpha but only by 1 point. Keep it in.

Overall, the HCAHPS reliably measures beliefs on providers and systems well but keep in mind, Alpha looks better with more items.

Exploratory Factor Analysis

Factor analysis “finds” factors in the data.



```
graph TD; A[Factor analysis “finds” factors in the data.] --> B[Because one question may not be enough, a factor analysis will help us know how well our questions work together.]; B --> C[For example, if you create a survey with 5 questions on confidence with medication, a factor analysis should identify those questions.]; C --> D[If similar survey questions are written well and tap into the same dimension, there should be high “factor loadings”.]
```

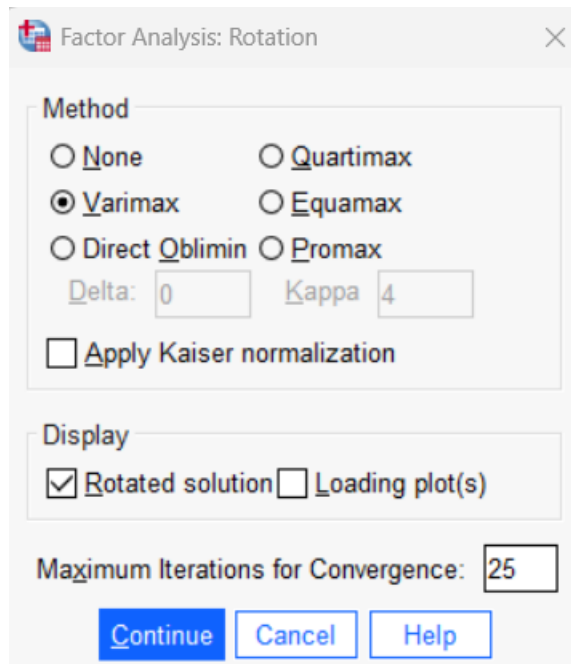
Because one question may not be enough, a factor analysis will help us know how well our questions work together.

For example, if you create a survey with 5 questions on confidence with medication, a factor analysis should identify those questions.

If similar survey questions are written well and tap into the same dimension, there should be high “factor loadings”.

Exploratory Factor Analysis: 2 factors identified from this survey

- You can accept most default settings
- Analyze -> Dimension Reduction -> Factor
- Rotation settings below



Factor Analysis: Rotation

Method

☐ None ☐ Quartimax

☒ Varimax ☐ Equamax

☐ Direct Oblimin ☐ Promax

Delta: 0 Kappa: 4

☐ Apply Kaiser normalization

Display

☒ Rotated solution ☐ Loading plot(s)

Maximum Iterations for Convergence: 25

Continue Cancel Help

2 factors identified:
Questions 2-15 are
highly correlated with
Factor 1.

Q26, Q27, and Q28
are highly correlated
with Factor 2.

We can now get factor scores or create our own average scores as the means of the 5 items and 3 items for each factor.

Rotated Component Matrix^a

	Component	
	1	2
q0002r	.893	.030
q0006r	.836	.079
q0008r	.883	.045
q0013r	.820	.036
q0015r	.907	.013
q0026r	-.021	.889
q0027r	.122	.899
q0028r	.012	.934

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax

Confirmatory Factor Analysis

Factor analysis “finds” factors in the data. Confirmatory Factor Analysis tests the “factor loadings” of predetermined factor variables.



Values range from -1 to 1. Factor loadings are “good” at 0.4/-0.4 or stronger.



Advantages of CFA: By isolating the shared variance of the multiple items from the unique variance, we’re able to obtain a better measure of the latent variable (i.e. factor).

Example: Factor Analysis

These values are factor loadings, the stronger the value, the stronger the correlation with each factor it is linked to.

CFA is a form of "Validity"

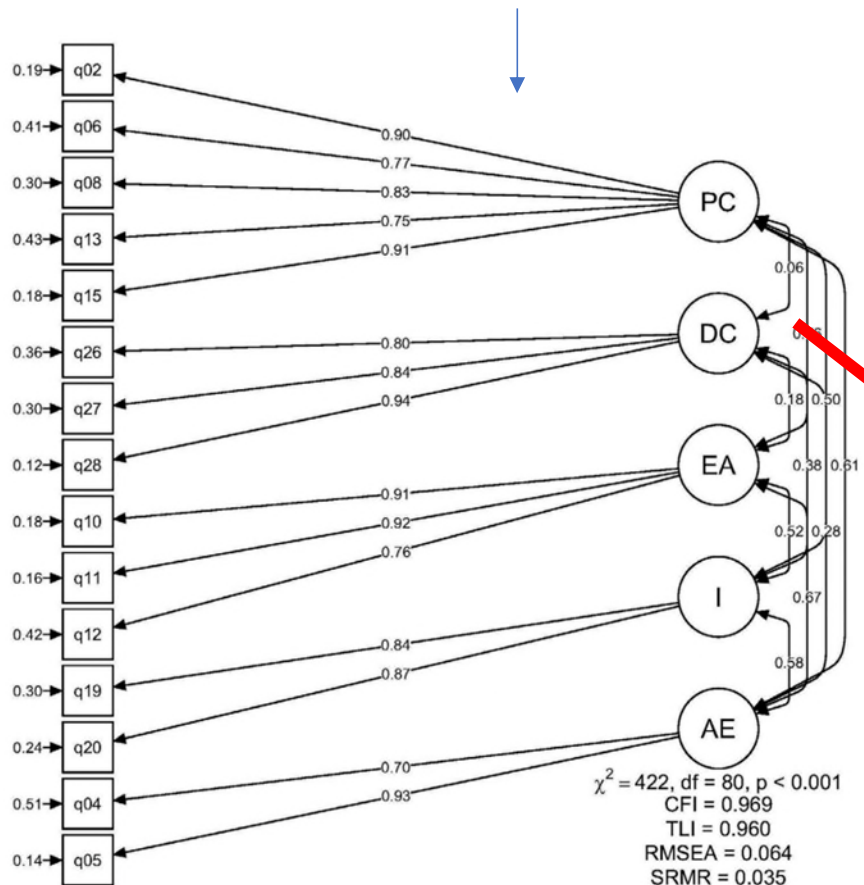


Table 3. EBP Readiness Assessment for Registered Nurses with Subscale Scores, Cronbach's Alpha, Survey Questions, and Confirmatory Factor Analysis (CFA)

Factor	Alpha ^a	No. ^b	Item	CFA
1. Practice Climate Subscale Score/SD 4.14 (0.87) N = 1,181	.92	2	Evidence-based nursing practice is important to me.	.90
		6	A journal club to discuss nursing research findings would be helpful.	.77
		8	Someone to assist with a literature search and obtain articles would increase use of evidence-based practices.	.83
		13	A bulletin board on my unit to share research articles would be helpful.	.75
		15	I am willing to try out new innovations found to be effective.	.91
2. Data Collection Subscale Score/SD 3.03 (1.08) N = 1,173	.90	26	I participate in the collection of data for research studies (i.e., conduct of research, not evidence-based practice projects).	.80
		27	I participate in the collection of data for quality improvement projects.	.84
		28	I participate in the collection of data for evidence-based practice projects.	.94
3. Evidence Appraisal Subscale Score/SD 3.76 (0.83) N = 1,178	.92	10	I can read a nursing research report and have a general notion about its strengths and weaknesses.	.91
		11	I can read a nursing research report and make a sound judgment about its scientific merit.	.92
		12	I am able to critique "synthesis" reports or technology assessments (e.g., systematic reviews) for a general understanding of their strengths and weaknesses.	.76
4. Implementation Subscale Score/SD 3.62 (0.90) N = 1,176	.85	19	I understand the process for implementing evidence into practice in my organization.	.84
		20	I am aware of effective strategies for implementing practice changes	.87
5. Access to Evidence Subscale Score/SD 3.75 (0.80) N = 1,181	.79	4	I have convenient access to nursing research journals.	.70
		5	I know where to find evidence (e.g., research findings or evidence-based clinical guidelines) to guide my practice.	.93

^aDenotes Cronbach's alpha (α) for each respective factor. Model fit indices reflected results of RMSEA (.063), CFI (.970), and SRMR (.034).

^bThe number (No.) in this column corresponds to the survey item number in the 2005 version.

Figure 1. Confirmatory factor analysis for full sample.

Summary

Survey Research Methods are a helpful approach in determining the reliability and validity of important things we examine.

Some of the Measurement methods we use are scales and factor analysis.

Factors allow us to safely collapse multiple items, improving the power of our analysis.

Survey methods tend to use only predictors, but our factors may be outcomes on their own.

R and Python generally work well with complex data but SPSS provides a more comprehensive set of options with survey data (e.g., recoding, factor analysis, etc.).



RN Continuing Education Hours -- Evaluation

This course is eligible for 1 CE contact hour. Please fill out the survey for 1 CE.

[2026 STATS with Steve Survey – Fill out form](#)

References

- Crawford, C., Rondinelli, J., Zuniga, S., Valdez, R., Cullen, L., Hanrahan, K., & Titler, M. (2020). “Testing of the Nursing Evidence-Based Practice Survey”. *Worldviews on Evidence-Based Nursing*, 118-128.

Appendix A

SPSS

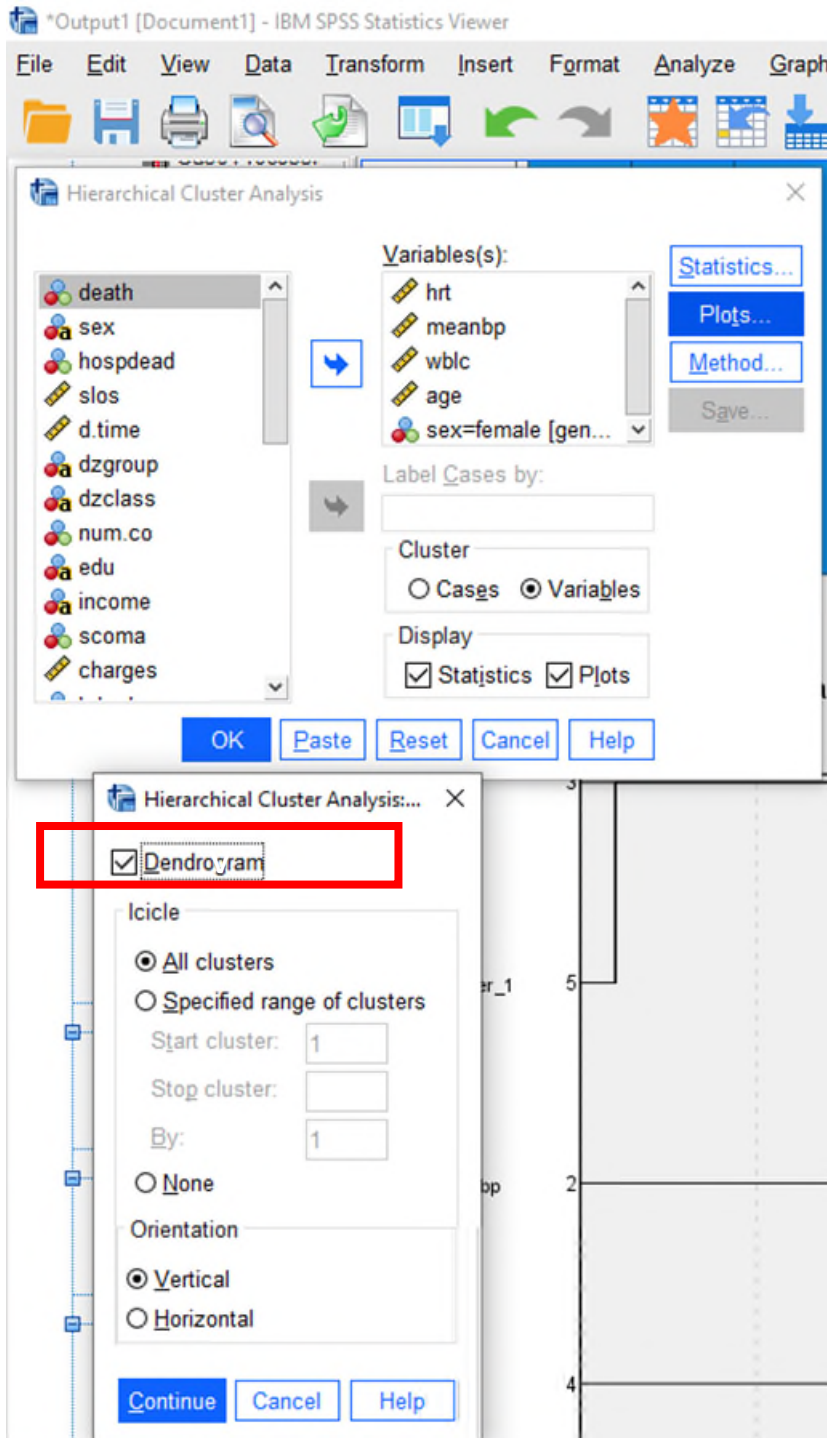
How to do a 1) Hierarchical Cluster Analysis, 2) Dendrogram, 3) Cronbach's alpha, and
4) Factor Analysis in SPSS

Hierarchical Cluster Analysis

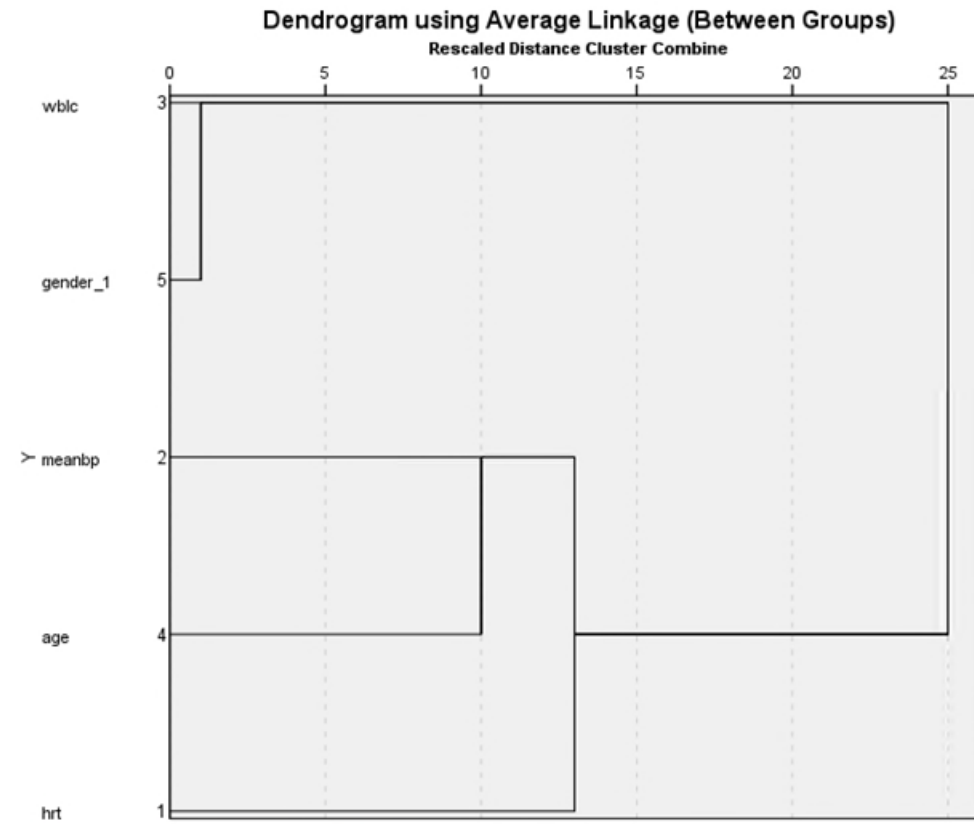
The screenshot displays the SPSS software interface. The top menu bar includes File, Edit, View, Data, Transform, Insert, Format, Analyze, Graphs, Utilities, Extensions, Window, and Help. The 'Analyze' menu is open, showing a list of statistical analysis options. The 'Classify' option is selected, which has opened a sub-menu. In this sub-menu, the 'Hierarchical Cluster...' option is highlighted with a red rectangular box. Other options in the 'Classify' sub-menu include TwoStep Cluster..., K-Means Cluster..., Cluster Silhouettes, Discriminant..., Nearest Neighbor..., ROC Curve..., and ROC Analysis....

On the left side of the interface, a project tree is visible, showing a hierarchy of folders and files. The 'Cluster' folder is expanded, revealing sub-items like Title, Notes, Centroid Linkage, Agglomerate, Horizontal Dendrogram, Proximities, Case Processing, and Average Linkage. Below the project tree, a vertical axis is labeled 'Number of clusters' with a scale from 2.0 to 4.0. To the right of this axis, a dendrogram is partially visible, showing the hierarchical clustering process for variables 'wblc' and 'gender_1'.

Hierarchical Cluster Analysis with a Dendrogram

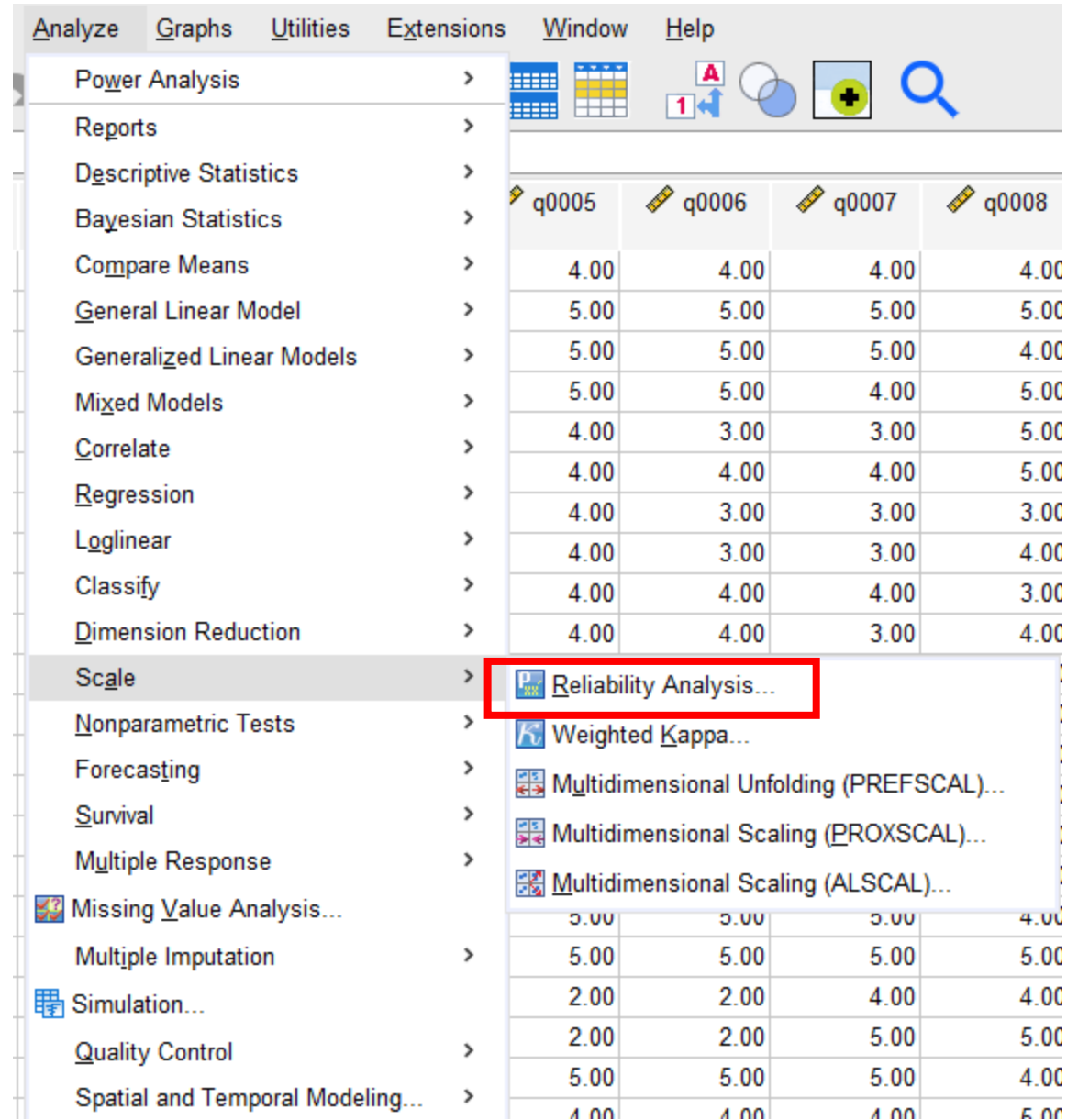


Dendrogram



Dataset Close D0.3045864166047004.

Cronbach's Alpha



The screenshot shows the SPSS 'Analyze' menu. The 'Scale' option is selected, and the 'Reliability Analysis...' sub-menu item is highlighted with a red rectangle. The 'Reliability Analysis...' item is preceded by a small icon of a document with a plus sign.

Menu structure:

- Analyze
 - Power Analysis
 - Reports
 - Descriptive Statistics
 - Bayesian Statistics
 - Compare Means
 - General Linear Model
 - Generalized Linear Models
 - Mixed Models
 - Correlate
 - Regression
 - Loglinear
 - Classify
 - Dimension Reduction
 - Scale**
 - Reliability Analysis...**
 - Weighted Kappa...
 - Multidimensional Unfolding (PREFSCAL)...
 - Multidimensional Scaling (PROXSCAL)...
 - Multidimensional Scaling (ALSCAL)...
 - Nonparametric Tests
 - Forecasting
 - Survival
 - Multiple Response
 - Missing Value Analysis...
 - Multiple Imputation
 - Simulation...
 - Quality Control
 - Spatial and Temporal Modeling...

	q0005	q0006	q0007	q0008
4.00	4.00	4.00	4.00	4.00
5.00	5.00	5.00	5.00	5.00
5.00	5.00	5.00	4.00	5.00
5.00	5.00	4.00	5.00	5.00
4.00	3.00	3.00	5.00	5.00
4.00	4.00	4.00	5.00	5.00
4.00	3.00	3.00	3.00	3.00
4.00	3.00	3.00	4.00	4.00
4.00	4.00	4.00	3.00	3.00
4.00	4.00	3.00	4.00	4.00
5.00	5.00	5.00	4.00	4.00
5.00	5.00	5.00	5.00	5.00
2.00	2.00	4.00	4.00	4.00
2.00	2.00	5.00	5.00	5.00
5.00	5.00	5.00	4.00	4.00
4.00	4.00	4.00	5.00	5.00

Reliability Analysis

Items:

- 22 I am actively pursuin...
- 23 I am planning to pur...
- 24 I am actively pursuin...
- 25 I am aware of eviden...
- 26 I participate in the c...
- 27 I participate in the c...
- 28 I participate in the colle...

Ratings:

Model: Alpha

Scale label:

OK Paste

Reliability Analysis: Statistics

Descriptives for

- ☒ Item
- ☒ Scale
- ☒ Scale if item deleted

Inter-Item

- ☐ Correlations
- ☐ Covariances

Summaries

- ☐ Means
- ☐ Variances
- ☐ Covariances
- ☐ Correlations

ANOVA Table

- ☒ None
- ☐ F test
- ☐ Friedman chi-square
- ☐ Cochran chi-square

Interrater Agreement: Fleiss' Kappa

- ☐ Display agreement on individual categories
- ☐ Ignore string cases
- ☒ String category labels are displayed in uppercase

Asymptotic significance level (%): 95

Missing

- ☒ Exclude both user-missing and system missing values
- ☐ User-missing values are treated as valid

Hotelling's T-square ☐ Tukey's test of additivity ☐

Intraclass correlation coefficient ☐

Model: Two-Way Mixed Type: Consistency

Confidence interval: 95 % Test value: 0

Continue Cancel Help

Cronbach's Alpha

Reliability Statistics

Cronbach's Alpha	N of Items
.896	3

Item-Total Statistics

	Scale Mean if Item Deleted	Scale Variance if Item Deleted	Corrected Item-Total Correlation	Cronbach's Alpha if Item Deleted
26 I participate in the collection of data for research studies (i.e., conduct of research, not evidence-based practice projects).	6.1867	4.971	.755	.886
27 I participate in the collection of data for quality improvement projects	5.8475	4.808	.787	.859
28 I participate in the collection of data for evidence-based practice projects	6.0500	4.811	.846	.809

Factor
Analysis

Analyze Graphs Utilities Extensions Window Help

Power Analysis >

Reports >

Descriptive Statistics >

Bayesian Statistics >

Compare Means >

General Linear Model >

Generalized Linear Models >

Mixed Models >

Correlate >

Regression >

Loglinear >

Classify >

Dimension Reduction >

Scale >

Nonparametric Tests >

Forecasting >

Survival >

Multiple Response >

 Missing Value Analysis...

Multiple Imputation >

 Simulation...

Quality Control >

Spatial and Temporal Modeling... >



q0005	q0006	q0007
4.00	4.00	4.00
5.00	5.00	5.00
5.00	5.00	5.00
5.00	5.00	4.00
4.00	3.00	3.00
4.00	4.00	4.00
4.00	3.00	3.00
4.00	3.00	3.00
4.00	4.00	4.00
Factor...		
Correspondence Analysis...		
Optimal Scaling...		
5.00	5.00	5.00
3.00	4.00	4.00
4.00	3.00	4.00
5.00	5.00	5.00
5.00	5.00	5.00
2.00	2.00	4.00
2.00	2.00	5.00
5.00	5.00	5.00
4.00	4.00	4.00

Factor Analysis

KFH SCPMG_Database_6.12.17.sav [DataSet2] - IBM SPSS Statistics Data Editor

File Edit View Data Transform Analyze Graphs Utilities Extensions Windows

Factor Analysis

Variables:

- 20 I am aware o...
- 21 I could find o...
- 22 I am actively...
- 23 I am plannin...
- 24 I am actively...
- 25 I am aware o...
- 29 I am able to ...
- In your opinion, ...
- In your opinion, ...
- Please select w...
- Please identiv i

Selection Variable:

Value...

OK Paste Reset Cancel Help

Factor Analysis: Rotation

Method

- ☐ None
- ☒ Varimax
- ☐ Direct Oblimin
- ☐ Quartimax
- ☐ Equamax
- ☐ Promax

Delta: 0 Kappa: 4

Display

☒ Rotated solution ☐ Loading plot(s)

Maximum Iterations for Convergence: 25

Continue Cancel Help

5.00	5.00	5
5.00	4.00	3
4.00	4.00	4
5.00	5.00	5
5.00	5.00	5
4.00	5.00	2
5.00	5.00	2
5.00	5.00	5
5.00	5.00	4
4.00	4.00	4
5.00	5.00	5
5.00	5.00	5
5.00	5.00	4
5.00	5.00	5
5.00	5.00	5

Factor
Analysis

Total Variance Explained

Component	Total	Initial Eigenvalues		Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
		% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	3.840	48.006	48.006	3.840	48.006	48.006	3.784	47.306	47.306
2	2.426	30.327	78.333	2.426	30.327	78.333	2.482	31.027	78.333
3	.415	5.187	83.520						
4	.389	4.857	88.376						
5	.313	3.914	92.290						
6	.258	3.219	95.510						
7	.192	2.403	97.913						
8	.167	2.087	100.000						

Extraction Method: Principal Component Analysis.

Rotated Component
Matrix^a

	Component	
	1	2
q0002r	.893	.030
q0006r	.836	.079
q0008r	.883	.045
q0013r	.820	.036
q0015r	.907	.013
q0026r	-.021	.889
q0027r	.122	.899
q0028r	.012	.934

Extraction Method: Principal

Appendix B

Hierarchical Cluster Analysis

1) Hierarchical Cluster Analysis, 2) Dendrogram

Analysis examples

- Hierarchical Cluster Analysis to identify variables that measure similar dimensions.
- Dendrogram to display Hierarchical Cluster Analysis results.

Hierarchical Cluster Analysis

Hierarchical Cluster Analysis allows us to see which variables (or patients) have similar response patterns.



Dendrograms let us view which variables are most correlated



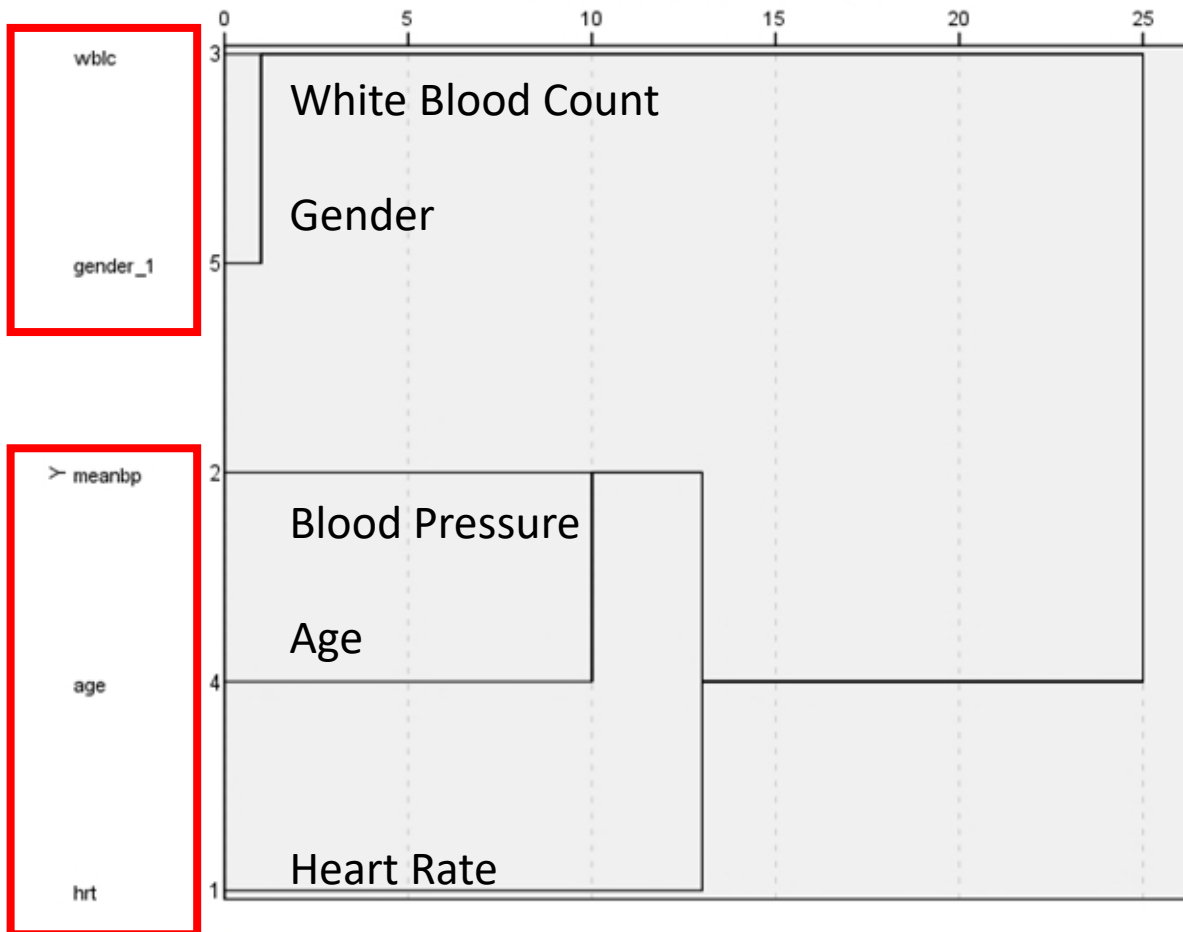
Scree plots can help us determine how many factors we have. We then use our HCA results to determine the factors.



You can create factors based on things like age, heart rate, blood pressure like we'll see later with survey questions.

Hierarchical Cluster Analysis and Dendrograms

Dendrogram using Average Linkage (Between Groups)
Rescaled Distance Cluster Combine



The dendrogram allows us to see which variables tend to have similar response patterns (i.e., correlated).

The HCA lets us detect factors such as the link between age, heart rate, and blood pressure.

For example, older patients tend to have lower heart rates.